

Aurora

Illuminating the
Future of Computing



Saanvi Subramanian, Kaustubh Grama
Innovation Plan

Bothell High School | 9130 NE 180th St, Bothell, WA 98011
February 14th, 2025

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I. Executive Summary

Problem:

Artificial intelligence is an innovative tool with the ability to revolutionize the world. However, the hardware needed to power this technology lacks scalability.

Current chips are simply not powerful enough to execute the complex computational demands of AI. Additionally, they require massive amounts of water to cool down and perform optimally, a practice that will cost the planet dearly if continued.



The question now persists:
How can we support the future of computing with efficient and sustainable technology?



Business Overview:

Aurora develops advanced photonic chips free from conventional electronic limitations through use of **light** instead of electrons for **faster**, more **efficient** and **scalable** computing.

Aurora offers three provisional-patented photonic chip solutions:

- **Ember**: entry-level, consumer-focused
- **Soleil**: mid-level, telecommunications applications
- **Halo**: flagship, high-range, specialized applications



Unique Value Proposition:

Aurora's patented, first-to-market *photonic computing* technology uses light particles instead of conventional electrons, offering up to **10x** the processing speed of leading chips like **NVIDIA's A-100** with less power needed.

Aurora
EMBER chip

- 6 TB/sec
- 5-10W



VS

Nvidia A-100
chip

- 600 GB/sec
- 250W



Customer Segmentation:

Aurora operates through a Business-to-Business (B2B) model, marketing to technology-related businesses detailed below:

- **PRIMARY**: Leading Artificial Intelligence Companies
- **SECONDARY**: US-based Semiconductor Producers
- **TERTIARY**: Corporate Technology Developers



OpenAI

Google AI

Qualcomm
MEDIATEK
BROADCOM NVIDIA



SAMSUNG

Solution:



Aurora creates computing chips using a 4-step **proprietary** approach that harnesses the power of light to provide a **scalable** and **sustainable** computing solution.

HOW OUR CHIPS WORK

STEP 1

Light Generation



Creates the optical signals

STEP 2

Light Manipulation



Changes the optical signals

STEP 3

Information Encoding



Imparts information in signals

STEP 4

Light Detection



Interprets the binary data in the signal

Investment and Return:



ASK: \$2,000,000 for 2.5% equity

AURORA VALUATION: \$80M

EXPECTED REVENUE IN 4 YRS: \$625M

INVESTOR RETURN IN 4 YRS: \$3,943,000

Investors can expect to see their money back (and more!) within the next four years, while **supporting the next generation of sustainable computing technologies with Aurora.**



II. Problem

Scalability ↔ Sustainability

The **scalability** and **sustainability** challenges of traditional chips are becoming increasingly apparent as AI models grow in size and complexity. The semiconductor industry has historically relied on **Moore's Law** – the observation that transistor density doubles roughly every two years – to deliver performance improvements. However, as we approach the **physical limits** of silicon-based transistor miniaturization (with features now just a few nanometers in size), this approach is yielding **diminishing returns** while requiring exponentially more power. Training a single large language model can now **consume** as much energy as hundreds of homes use in a year, and the **carbon footprint** of these computations is becoming a **serious environmental concern**. The current trajectory of power consumption in computing, if based on traditional chip architectures, is simply **not sustainable** from both an **economic** and **environmental** perspective.



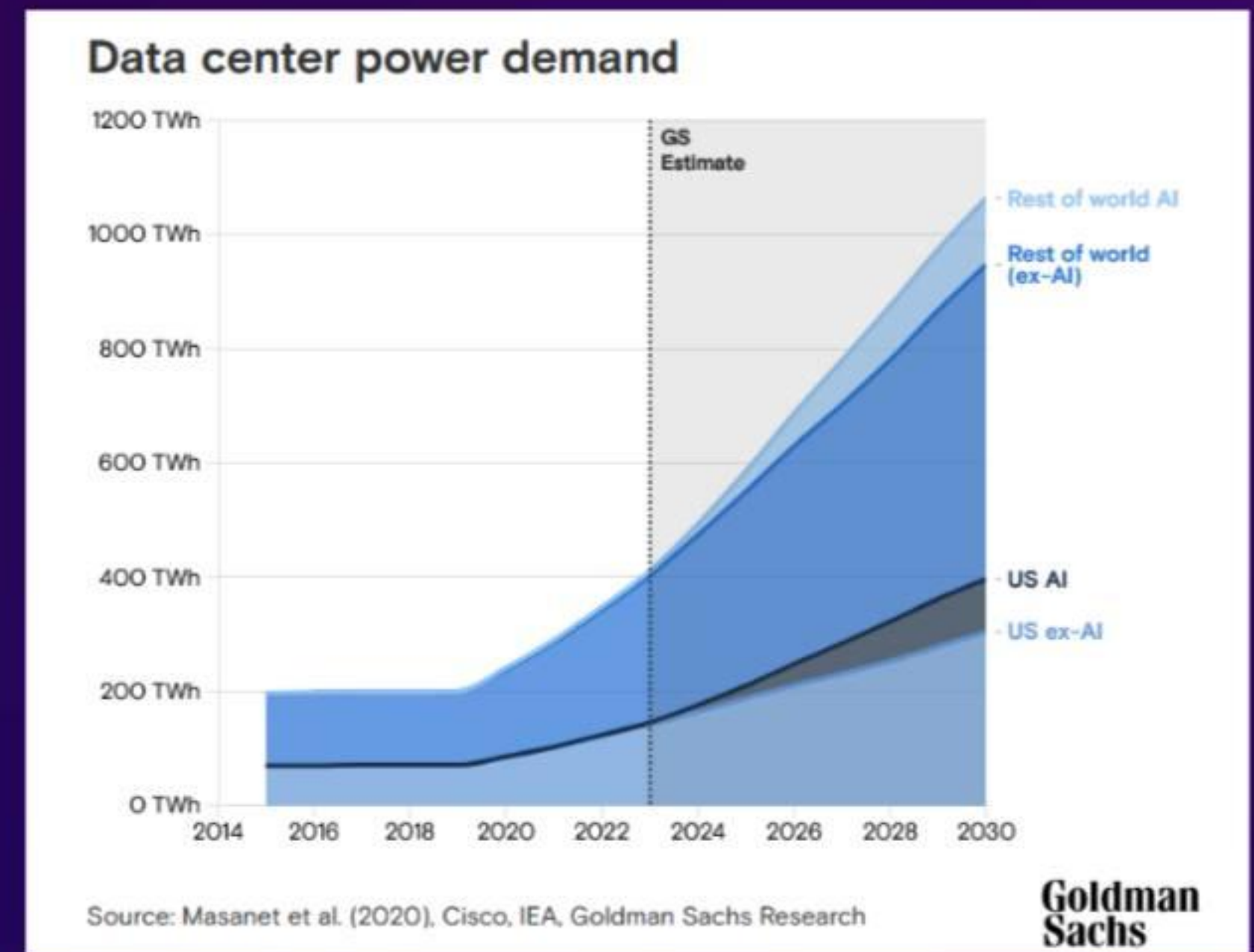
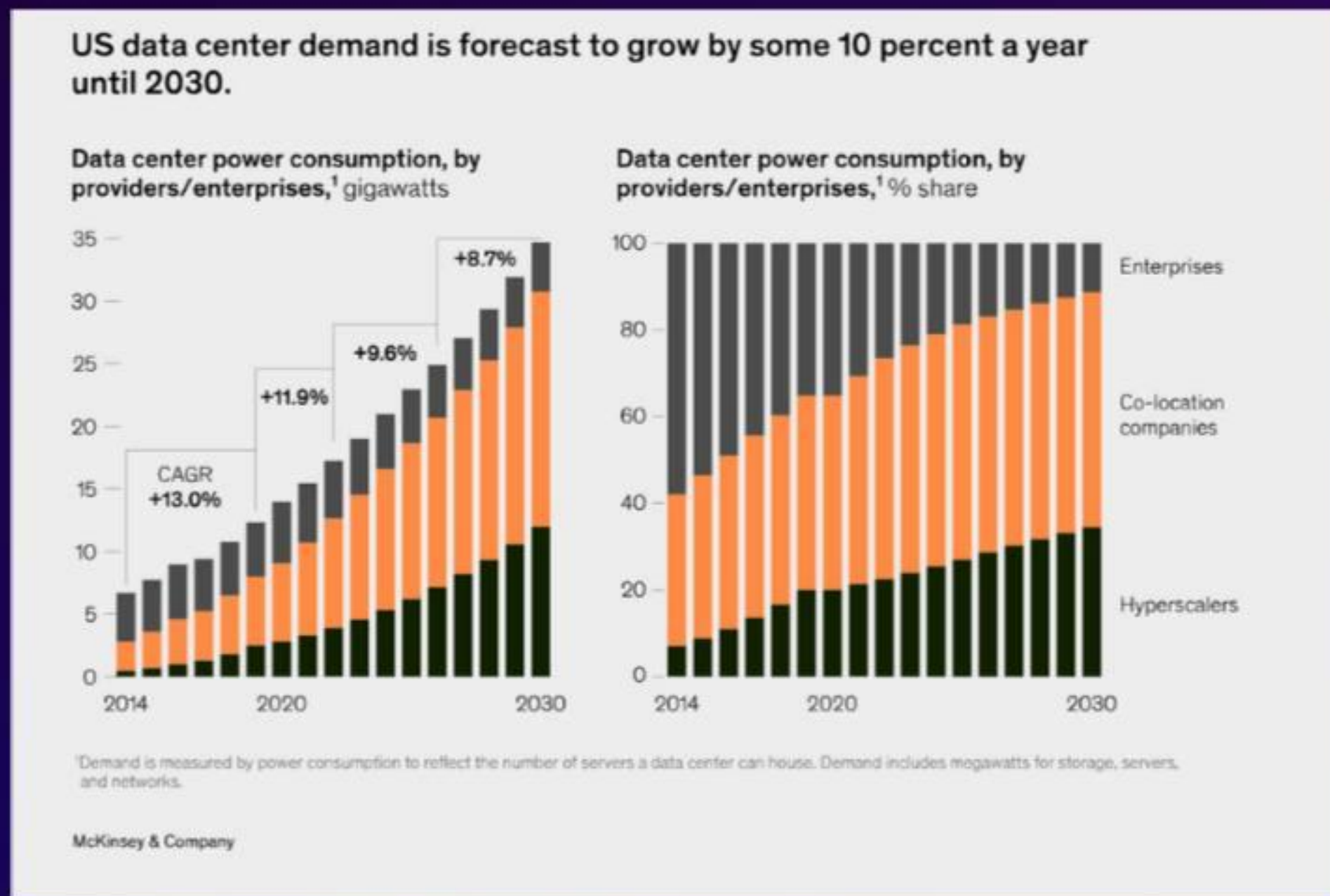
Current **cooling solutions** for these chips rely heavily on conventional methods like heat sinks, fans, and liquid cooling systems, which themselves **consume significant power** and add to the overall **energy burden**. For instance, **data centers** running AI workloads typically require massive air conditioning systems that can consume almost as much power as the computing hardware itself. Liquid cooling solutions, while more efficient than air cooling, still face fundamental **limitations** in **heat transfer** capacity and require complex plumbing systems that add to infrastructure costs. The **thermal resistance** between the chip's internal layers and the cooling solution creates "**hot spots**" that can lead to **thermal throttling**, where the chip must **reduce its performance to prevent damage**, further reducing computational efficiency. AI's environmental impact is expected to **escalate** considerably, with the global AI energy demand projected to exponentially increase to at least **10 times** the current level and exceed the annual electricity consumption of a small country like Belgium by 2026. In the United States, the rapidly growing AI demand is poised to drive data center energy consumption to about 6% of the nation's total electricity usage in 2026, adding further pressure on grid infrastructures and highlighting the **urgent need for sustainable solutions** to support continued AI advancement.



These cooling challenges become even more pronounced in **large-scale AI training facilities**, where **thousands of chips** might be running simultaneously. The **heat density** in these facilities can exceed 50 kilowatts per rack, pushing the limits of what traditional cooling technologies can handle. The industry's current approach of simply adding more cooling capacity is creating a **vicious cycle**: more powerful chips require more cooling, which requires more energy, which generates more heat, requiring even more cooling. This unsustainable spiral is reflected in the fact that major AI training runs can now consume **millions of kilowatt-hours of electricity**, with some estimates suggesting that a single large language model training session can generate a carbon footprint equivalent to the **lifetime emissions of five average American cars**.



AI model builders have been making AI algorithms smarter by either increasing the size of the model or by increasing the amount of data required to train them. However, the growth in the required computation is at a scale that traditional electronics cannot catch up with.



III. Customer Segments

We market Aurora's technologies to companies, in a business-to-business (B2B) model. We identify three key customer segmentations for Aurora's photonic computing chips below.

Primary Segment: Leading Artificial Intelligence Companies

Aurora's primary customer segment includes leading companies that have made **Artificial Intelligence** technologies, and those who prioritize it in their **consumer products**. Our goal is that these companies switch their existing chipset infrastructure with Aurora's novel approach of using **photonics** to relay information, allowing for an increasingly **optimized** technological and **energy** approach. These companies are predicted to exhibit **rapid growth**. The benefits of Aurora computing chips lie in its ability to power **scalable, capable** futures for **consumer oriented** Artificial Intelligence solutions, boosting enterprise **profits** and **innovation** for decades to come.



Secondary Segment: US-based Semiconductor Producers

Aurora's secondary customer segment includes companies that are already in the business of mass producing **semiconductors** and chips for a vast array of applications. Major players in the US-based semiconductor industry are located in California's Silicon Valley and include **NVIDIA**, **Qualcomm**, and **Intel**. These companies can use our technology to advance their own products, such as **cell phones, televisions, and computers**. Aurora will revamp the innovation landscape by reaching out to existing entities in the semiconductor industry so that everyone can experience the next generation of computing.



Tertiary Segment: Corporate Technology Developers

Aurora's tertiary customer segment includes technology developers, such as Apple, Google, Oracle, and Salesforce. These companies are pushing the boundaries of innovation in their respective niches, and we at Aurora want to be a part of that. By marketing our product to these organizations, we can increase integration of photonic computing in a wide array of consumer technologies.



IV. Unique Value Proposition



Photonic computing chips offer several fundamental **advantages** that directly address the limitations of traditional electronic chips by using **light** instead of electrons to process and transmit information. In photonic chips, light waves can pass through each other **without interference**, enabling **true parallel processing** through the same physical space—something impossible with electronic signals. This dramatically reduces the energy needed for computation since **photons require virtually no energy to travel through waveguides**, unlike electrons moving through copper traces which face resistance and generate heat.

The **thermal advantages** of photonic computing are particularly significant. Since photonic operations **generate minimal heat** compared to electronic transistors, they **drastically reduce cooling requirements**. Photonic chips can operate at **much higher frequencies** (potentially reaching the terahertz range) **without the thermal constraints** that plague electronic systems. This higher operating frequency translates to faster processing speeds without the corresponding increase in heat generation that would occur in traditional electronic chips. Additionally, because light beams can carry multiple signals simultaneously through wavelength division multiplexing (WDM), a single photonic waveguide can replace multiple electronic interconnects, further reducing power consumption and heat generation.

When applied specifically to AI workloads, photonic computing offers **unique capabilities** for performing simultaneous operations—the fundamental building blocks of artificial intelligence—with **unprecedented efficiency**. Complex computations can be performed nearly **instantaneously** using optical interference patterns, dramatically **accelerating** AI training and inference while consuming orders of magnitude less power than electronic alternatives. This architectural advantage, combined with the inherent energy efficiency and thermal benefits of photonics, creates a path toward **sustainable scaling** of AI computing power that simply isn't possible with traditional electronic architectures.

We outlined a SWOT analysis for this new technology to examine its strengths, weaknesses, opportunities in the market, and potential threats.

Advantages of Photonic Computing

Speed: Photons travel at the speed of light, enabling optical computing systems to perform computations much faster than electronic counterparts

Energy Efficiency: Optical computing consumes less energy than electronic computing since photons experience minimal resistance and do not generate heat during data transmission

Parallelism: Light-based systems inherently support parallel processing, allowing multiple computations to be performed simultaneously, leading to significant performance gains for AI tasks

Scalability: Optical components can be integrated densely on a chip, enabling the development of compact and scalable optical computing architectures suitable for AI hardware



Opportunities and Potential Applications

Deep Learning: Optical computing can accelerate training and inference tasks for deep neural networks, enabling faster model training and real-time AI applications

Image and Video Processing: Optical computing is well-suited for tasks such as image recognition, object detection, and video analysis due to its high-speed processing capabilities

Natural Language Processing: Optical computing can enhance the performance of natural language processing algorithms by enabling rapid text analysis and translation

Sensor Data Processing: Optical computing can process large volumes of sensor data from IoT devices and environmental sensors in real-time, facilitating data-driven decision-making

Quantum Computing: Optical computing techniques are also being explored for quantum computing applications, promising breakthroughs in AI algorithms and optimization problems



Challenges and Considerations

Integration Complexity: Integrating optical components with existing electronic hardware poses technical challenges related to compatibility, alignment, and signal conversion

Manufacturing Costs: Optical components such as lasers and photodetectors can be expensive to manufacture, hindering the widespread adoption of optical computing technology

Signal Noise: Optical signals are susceptible to noise from environmental factors such as temperature fluctuations and optical imperfections, requiring robust error-correction techniques



V. Solution

Aurora provides a variety of products using our patent-pending technology. Through our proprietary four-step workflow, our photonic-powered chips deliver unparalleled speed, reliability, and efficiency in computing solutions.

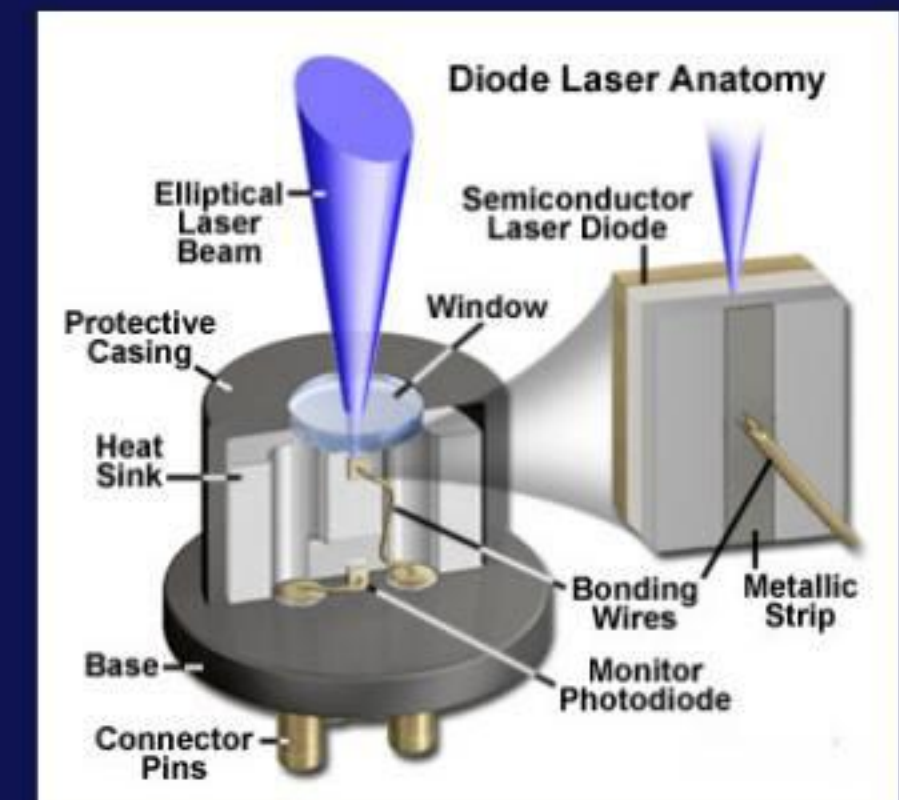
How our chips operate

STEP 1

Light Generation



- **Laser Source**
 - Chips use photons instead of electrons, so we integrate a **laser diode (a tiny semiconductor device)** onto each chip.
 - When a current is applied to the diode, it emits a beam of coherent light, meaning all the waves are coordinated with each other for maximum precision.

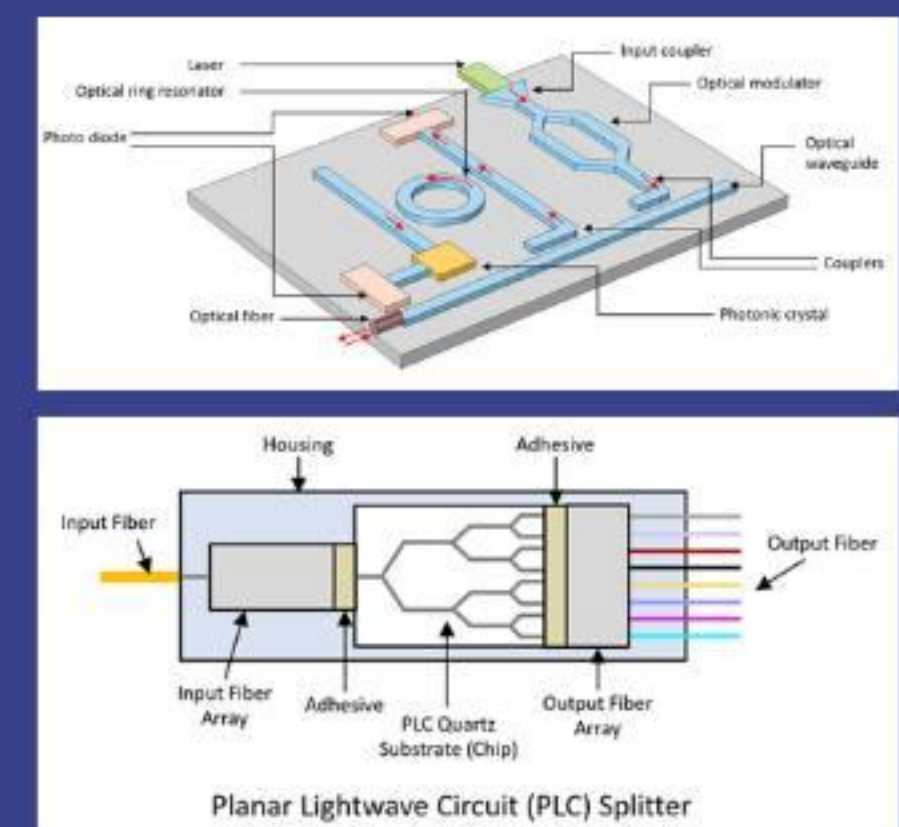


STEP 2

Light Manipulation



- **Waveguides:** Channels made of silicon nitride direct and confine light within the chip
- **Splitters:** Divide light into multiple paths for parallel processing.
- **Filters:** Select specific wavelengths to ensure only relevant light is processed.
- **Couplers:** Combine/separate light beams for complex photonic circuits.

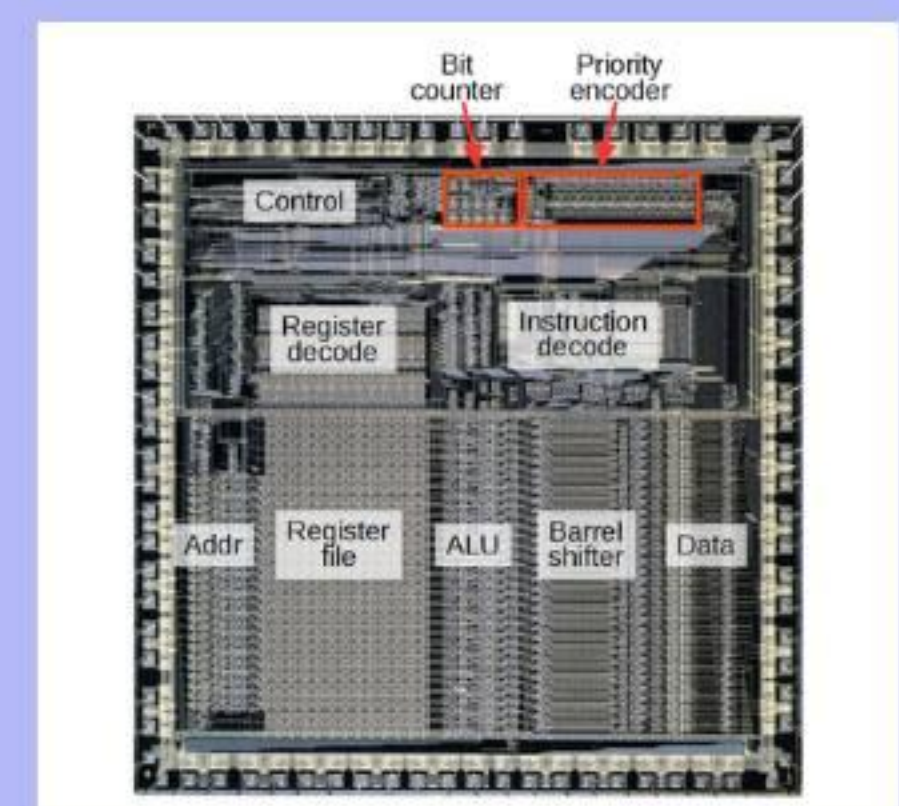


STEP 3

Information Encoding



- **Amplitude Modulation:** Varies light intensity to encode binary data (0s and 1s).
- **Phase Shift Keying:** Shifts the position of the light wave to represent information.
- **Polarization Shift Keying:** Alters the orientation of light oscillation for encoding data.

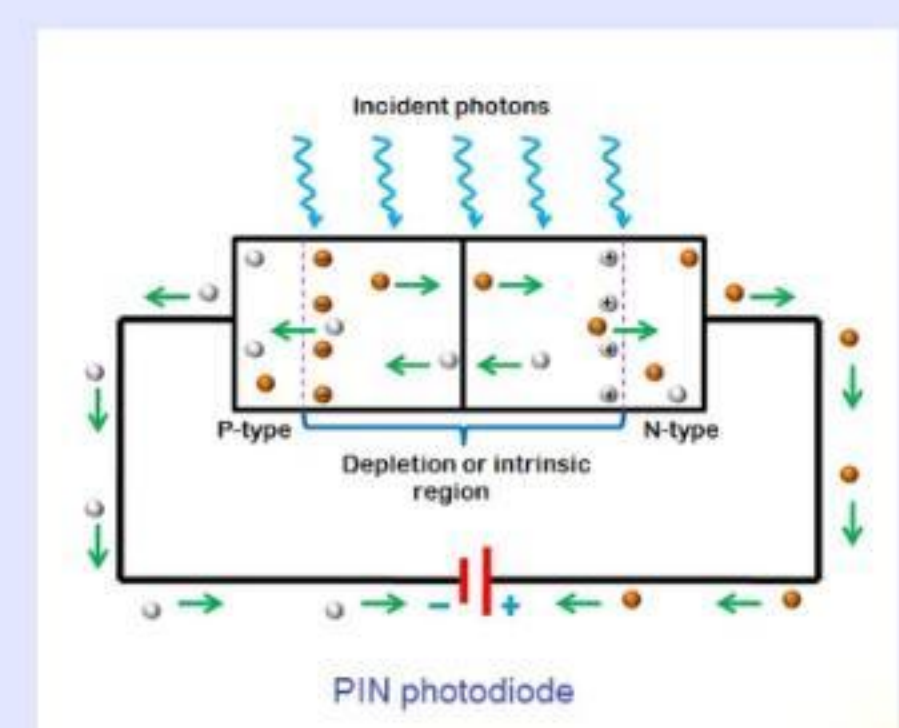


STEP 4

Light Detection



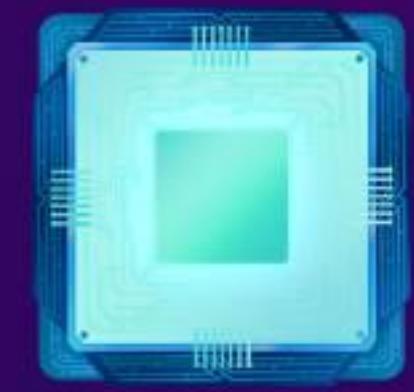
- **Photodetectors:** Uses Electron-hole Pair Generation to convert absorbed light into electric current.
- **Photodiodes:** Generates currents proportional to beam intensity, carrying encoded information.
- **Photomultiplier Tubes:** Amplifies light into measurable electric charges using the Photoelectric Effect.



Product Offerings

Ember

How it works: Single-core photonic processing unit with silicon photonics, Basic wavelength division (4 channels), Hybrid electronic-photonic design with integrated laser source, Standard silicon waveguides for light transmission



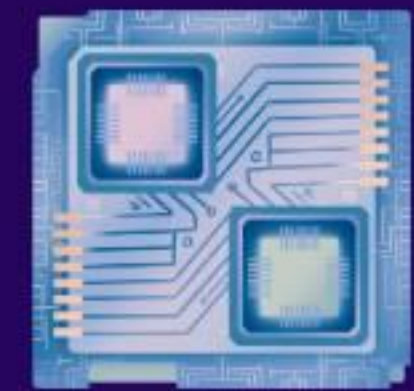
Applications: IoT devices and sensors, Computer vision in consumer devices, Small-scale machine learning tasks

Features: Processing speed: 100 TOPS (Tera Operations Per Second), Power consumption: 5-10 W, Integrated photonic memory buffer: 2GB, Basic computing acceleration

Cost Structure: Manufacturing cost: \$40-\$60 per unit, Retail price: \$150 - \$200, Volume pricing available for purchases of 1000+ units, Target margin: 60 - 65%

Soleil

How it works: Dual-core photonic processing architecture, Advanced WDM with 16 channels, Dedicated photonic memory interfaces, Proprietary low-loss waveguide technology, Integrated optical switching matrix



Applications: Data center acceleration, Real-time AI inference, Scientific computing, Financial modeling, Medical imaging processing

Features: Processing speed: 500 TOPS, Power consumption: 20-30W, Integrated photonic memory: 8GB, Advanced error correction, Temperature range: -10°C to 85°C, Form factor: 35mm x 35mm, Multi-chip scaling support

Cost Structure: Manufacturing cost: \$200-250 per unit, Retail price: \$800-1000, Volume pricing for 500+ units, Target margin: 70-75%

Halo

How it works: Quad-core photonic processing architecture, Ultra-dense WDM (32+ channels), Full photonic interconnect fabric, Integrated frequency comb source, Advanced phase control systems, Quantum-compatible optical circuits



Applications: Large language model training, High-performance computing, Quantum computing interface, Advanced AI research, Real-time complex simulation

Features: Processing speed: 2000+ TOPS, Power consumption: 50-70W, Integrated photonic memory: 32GB, Advanced error correction and redundancy, Temperature range: -20°C to 95°C, Form factor: 50mm x 50mm, Multi-rack scaling capability, Real-time monitoring and optimization

Cost Structure: Manufacturing cost: \$800-1000 per unit, Retail price: \$3000-4000, Custom pricing for data center deployments, Target margin: 75-80%

All three of our advanced chip offerings share common advantages that make any one of them a great fit for various levels of computing demand. Since they are all powered using photonic computing, they boast significantly reduced heat generation compared to electronic chipsets. As a result, they have lower requirements for cooling infrastructure and better energy efficiency. They also have much faster data movement and future-proof architecture for supporting upgrades to meet the ever-expanding capabilities of artificial intelligence. They also are in-built with software compatibility with major AI frameworks in the existing marketplace.

VI. Conclusion

The fundamental limitations of traditional electronic computing have become increasingly apparent as AI workloads grow more **demanding**. Current chips, based on moving electrons through silicon, face severe **thermal and power constraints**. The enormous energy requirements of AI computation, combined with **inefficient cooling solutions** that themselves consume significant power, have created an **unsustainable** cycle where data centers consume millions of kilowatt-hours of electricity and generate **massive carbon footprints**. Traditional cooling methods, whether air-based or liquid, struggle to manage heat densities that can exceed 50 kilowatts per rack, leading to **thermal throttling** and **reduced performance**.

Photonic computing emerges as a transformative **solution** to these challenges by utilizing **light** instead of electrons for computation and data movement. By enabling **true parallel processing** through wavelength division and generating minimal heat during operation, photonic chips can dramatically **reduce** both **power consumption** and **cooling requirements**. The ability to perform operations through optical interference patterns makes them particularly well-suited for **AI workloads**, while their capacity for high-frequency operation **without thermal constraints** enables **unprecedented processing speeds**. The technology's inherent **advantages** in data movement **efficiency**, with the ability to transmit multiple signals **simultaneously** through a single waveguide, directly addresses the energy and thermal bottlenecks that plague traditional electronic systems.

Our **innovative three-tiered approach** to photonic computing solutions – **Ember, Soleil, and Halo** – provides a comprehensive pathway for organizations to adopt this revolutionary technology. Starting with the entry-level **Ember** at \$150–200, designed for AI and IoT applications with its single-core architecture and 100 TOPS processing speed, organizations can begin their transition to photonic computing. The mid-range **Soleil**, priced at \$800–1000, offers enhanced capabilities with its dual-core design and 500 TOPS processing power, perfect for data center acceleration and professional computing needs. At the enterprise level, the **Halo** chipset, priced at \$3000–4000, delivers extraordinary performance with its quad-core architecture and 2000+ TOPS processing speed, making it ideal for large language model training and quantum computing interfaces.

Investment Ask

To **accelerate** our go-to-market strategy and scale production, we are seeking an **investment** of **\$2 million** for a **2.5% equity stake** in the company. These funds will be allocated toward **finalizing** development, **expanding** manufacturing capacity, and **driving** strategic partnerships with industry leaders like Apple, Google, and Salesforce. By partnering with us, you join a **transformative** journey to **redefine** computing and unlock **substantial returns** in a rapidly expanding market. We aim to catalyze a **new era of innovation**, ensuring that photonic computing becomes the cornerstone of the next wave of technological progress. **Together, we can illuminate a sustainable, scalable, and transformative future in computing technologies.**

Return on Investment (ROI) Calculations - \$M

Metric	Year 1 – 2025	Year 2 – 2026	Year 3 – 2027	Year 4 – 2028	Total
Total Revenue	25	100	200	300	625
Net Profit	3.75	20	50	84	157.75
ROI	0.093	0.5	1.25	2.1	3.94

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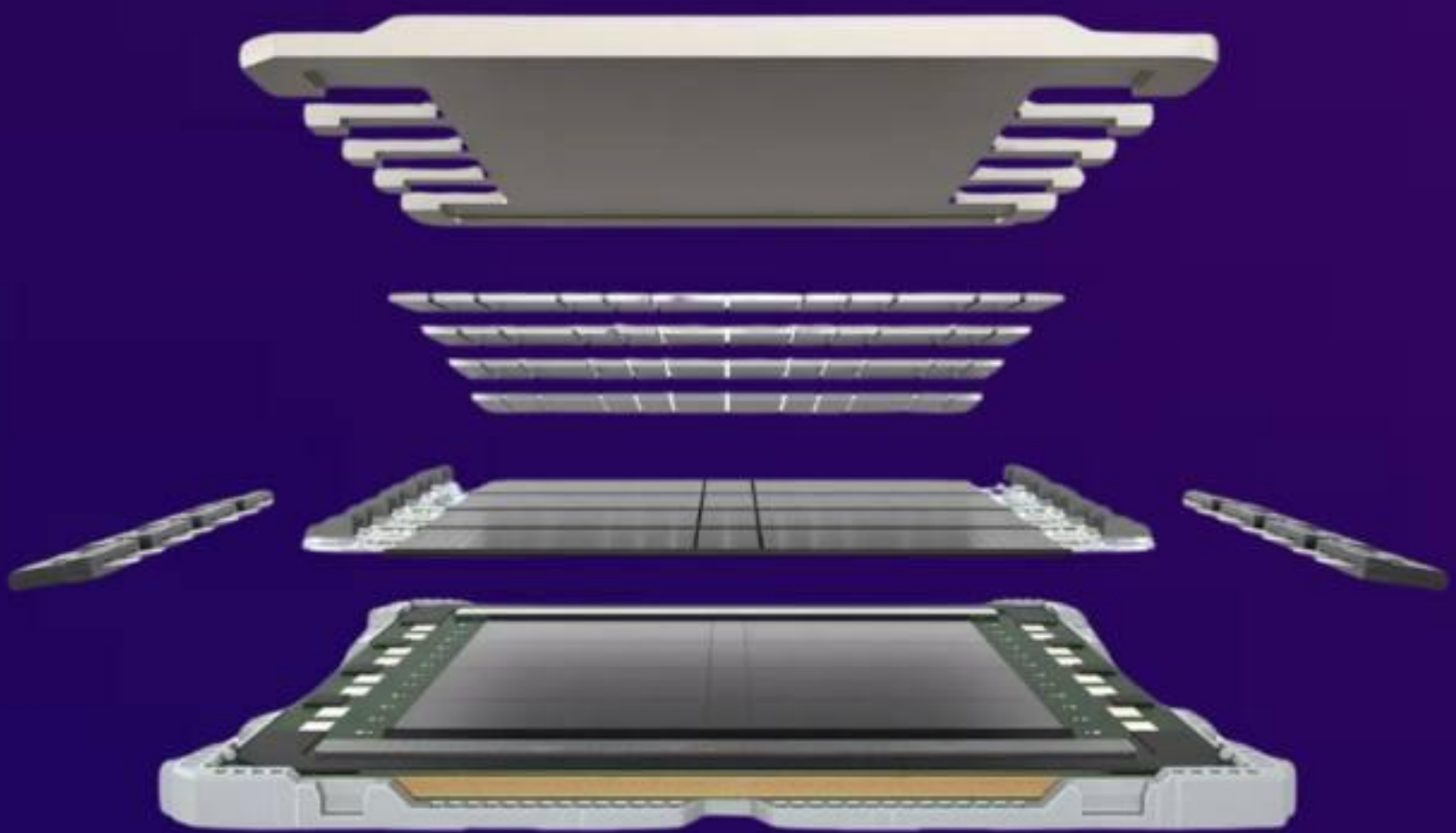
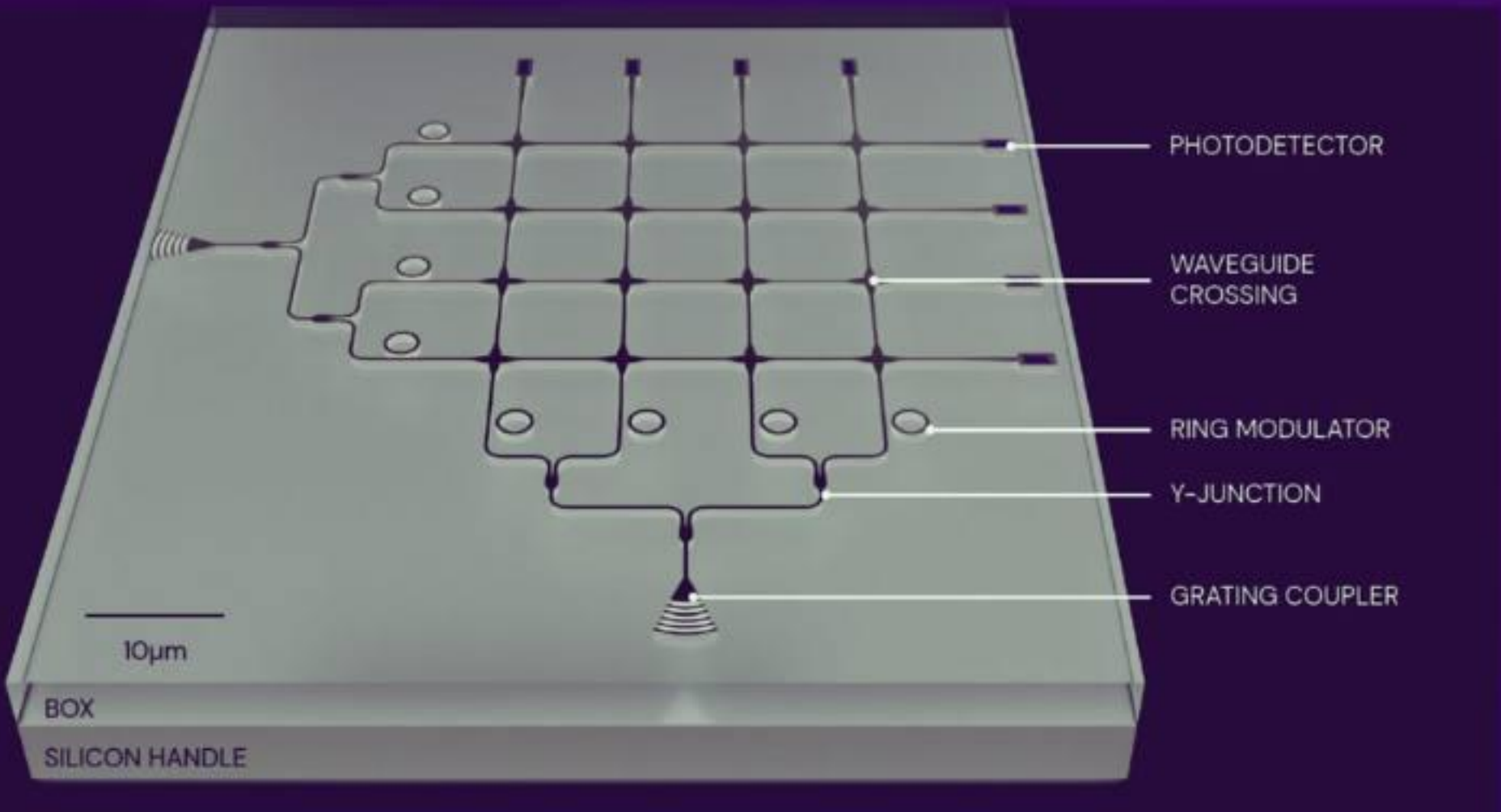
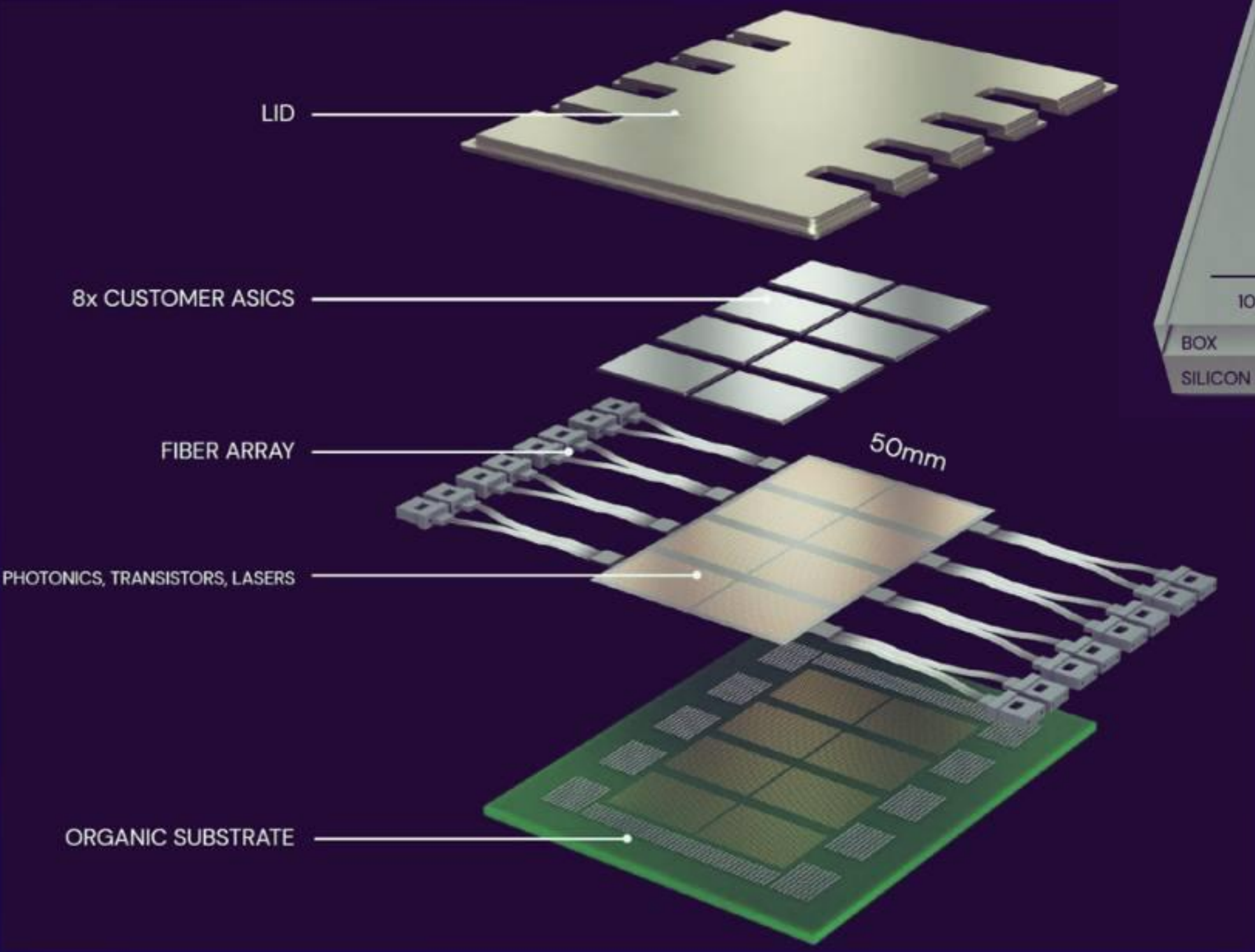
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VIII. Appendix

Sample Chip Component Layouts



Chip Anatomy



Proof of Patents

US 20200284984A1

(19) **United States**
(12) **Patent Application Publication**

(10) Pub. No.: US 2020/0284984 A1
(43) Pub. Date August 20, 2024

(54) **SYSTEM FOR PHOTONIC COMPUTING**

(71) Applicant: Aurora, Inc.
(72) Inventor: Saanvi Subramanian

(21) Appl. No.: 16/848,429
(22) Filed: Aug. 20, 2024

Publication Classification

(51) Int. Cl. G02B 6/293 (2006.01)
G02B 6/42 (2006.01)
G06E 3/00 (2006.01)
(52) U.S. CL. G02B 6/29355 (2013.01); H04B 10/70 (2013.01); G06E 3/008 (2013.01); G02B 6/4215 (2013.01)

(57) **ABSTRACT**
A system for photonic computing, preferably including an input module, computation module, and/or control module, wherein the computation module preferably includes one or more filter banks and/or detectors. A photonic filter bank system, preferably including two waveguides and a plurality of optical filters optically coupled to one or more of the waveguides. A method for photonic computing, preferably including controlling a computation module, controlling an input module, and/or receiving outputs from the computation module.

Related U.S. Application Data

(62) Division of application No. 16/672,231, filed on Nov. 1, 2019, now Pat. No. 10,656,336.
(60) Provisional application No. 62/757,647, filed on Nov. 8, 2018.

The block diagram shows a system 100 with an input module 110, a computation module 120, and a control module 130. The input module 110 is connected to the computation module 120, which is connected to the control module 130. The computation module 120 is further detailed with a laser array, modulators, weight banks, and detectors, all connected to the control module 130.

